



主观题 HW6

HW 6.1

在雨课堂的填空题中，选择一个普遍有效的公式给出证明，并选择一个不是普遍有效的公式举出反例。（10分）

证(3):

$$\begin{aligned}
 & ((\exists x)P(x) \rightarrow (\forall x)Q(x)) \rightarrow (\forall x)(P(x) \rightarrow Q(x)) \\
 &= \neg((\forall x)\neg P(x) \vee (\forall x)Q(x)) \vee (\forall x)(\neg P(x) \vee Q(x)) \\
 &= ((\exists x)P(x) \wedge (\exists x)\neg Q(x)) \vee (\forall x)(\neg P(x) \vee Q(x)) \\
 &= \neg((\forall x)\neg P(x) \vee (\forall x)Q(x)) \vee (\forall x)(\neg P(x) \vee Q(x)) \\
 &= ((\forall x)\neg P(x) \vee (\forall x)Q(x)) \rightarrow (\forall x)(\neg P(x) \vee Q(x)) \\
 &= T
 \end{aligned}$$

给(1)举反例:

在(1)(2)域上研究:

$$P(1)=P(2)=Q(1)=F, Q(2)=T;$$

$$\begin{aligned}
 & \text{原式} = ((P(1) \leftrightarrow Q(1)) \vee (P(2) \leftrightarrow Q(2))) \rightarrow ((P(1) \vee P(2)) \leftrightarrow (Q(1) \vee Q(2))) \\
 &= T \rightarrow F \\
 &= F
 \end{aligned}$$

解:

HW 6.2

证明下列等值式和蕴含式:

$$(1) (\forall x)(P(x) \vee q) \rightarrow (\exists x)(P(x) \wedge q) = ((\exists x)\neg P(x) \wedge \neg q) \vee ((\exists x)P(x) \wedge q) \quad (5 \text{ 分})$$

$$(2) (\forall y)(\exists x)((P(x) \rightarrow q) \vee S(y)) = ((\forall x)P(x) \rightarrow q) \vee (\forall y)S(y) \quad (5 \text{ 分})$$

$$(3) (\forall x)P(x) \rightarrow q = (\exists x)(P(x) \rightarrow q) \quad (5 \text{ 分})$$

$$(4) (\exists x)(P(x) \rightarrow Q(x)) = (\forall x)P(x) \rightarrow (\exists x)Q(x) \quad (5 \text{ 分})$$

$$(5) (\exists x)P(x) \rightarrow (\forall x)Q(x) \Rightarrow (\forall x)(P(x) \rightarrow Q(x)) \quad (5 \text{ 分})$$

$$(6) (\exists x)P(x) \wedge (\forall x)Q(x) \Rightarrow (\exists x)(P(x) \wedge Q(x)) \quad (5 \text{ 分})$$

$$(7) ((\forall x)P(x) \wedge (\forall x)Q(x) \wedge (\exists x)R(x)) \vee ((\forall x)P(x) \wedge (\forall x)Q(x) \wedge (\exists x)S(x)) = (\forall x)(P(x) \wedge Q(x)) \wedge (\exists x)(R(x) \vee S(x)) \quad (5 \text{ 分})$$

$$\begin{aligned}
 (1) \text{左式} &= (\forall x)P(x) \vee q \rightarrow (\exists x)P(x) \wedge q \\
 &= \neg((\forall x)P(x) \vee q) \vee ((\exists x)P(x) \wedge q) \\
 &= ((\exists x)\neg P(x) \wedge \neg q) \vee ((\exists x)P(x) \wedge q) = \text{右式}
 \end{aligned}$$

$$\begin{aligned}
 (2) \text{右式} &= (\neg((\forall x)P(x)) \vee q) \vee (\forall y)S(y) \\
 &= ((\exists x)\neg P(x) \vee q) \vee (\forall y)S(y)
 \end{aligned}$$

$$\begin{aligned}
&= (\forall y)((\exists x)\neg P(x) \vee q \vee S(y)) \\
&= (\forall y)(\exists x)(\neg P(x) \vee q \vee S(y)) \\
&= (\forall y)(\exists x)((P(x) \rightarrow q) \vee S(y)) = \text{左} \\
&\text{式}
\end{aligned}$$

$$(3) \text{左式} = \neg((\forall x)P(x)) \vee q$$

$$= (\exists x)\neg P(x) \vee q$$

$$= (\exists x)(\neg P(x) \vee q)$$

$$= (\exists x)(P(x) \rightarrow q) = \text{右式}$$

$$(4) \text{左式} = (\exists x)(\neg P(x) \vee Q(x))$$

$$= (\exists x)\neg P(x) \vee (\exists x)Q(x)$$

$$= \neg((\exists x)\neg P(x)) \rightarrow (\exists x)Q(x)$$

$$= (\forall x)P(x) \rightarrow (\exists x)Q(x) = \text{右式}$$

$$(5) \text{左式} = \neg((\exists x)P(x)) \vee (\forall x)Q(x)$$

$$= (\forall x)\neg P(x) \vee (\forall x)Q(x)$$

$$\text{右式} = (\forall x)(\neg P(x) \vee Q(x))$$

$$\text{左式} \Rightarrow \text{右式}$$

(6)若前项为真

即 $(\exists x)P(x)$ 为真且 $(\forall x)Q(x)$ 为真

设 $P(x_0)$ 为真且 $Q(x_0)$ 为真

则 $P(x_0) \wedge Q(x_0)$ 为真

即后项 $(\exists x)(P(x) \wedge Q(x))$ 为真

所以 $(\exists x)P(x) \wedge (\forall x)Q(x) \Rightarrow (\exists x)(P(x) \wedge Q(x))$

$$(7) \text{左式} = ((\forall x)P(x) \wedge (\forall x)Q(x) \wedge (\exists x)R(x)) \vee ((\forall x)P(x) \wedge (\forall x)Q(x) \wedge (\exists x)S(x))$$

$$= ((\forall x)(P(x) \wedge Q(x)) \wedge (\exists x)R(x)) \vee ((\forall x)(P(x) \wedge Q(x)) \wedge (\exists x)S(x))$$

$$= (\forall x)(P(x) \wedge Q(x)) \wedge ((\exists x)R(x) \vee (\exists x)S(x))$$

$$= (\forall x)(P(x) \wedge Q(x)) \wedge (\exists x)(R(x) \vee S(x)) = \text{右式}$$

证明:

HW 6.3

求下列 (1) 到 (5) 的前束范式, 以及 (6) 和 (7) 的 Skolem 标准型, 即仅保留全称量词的前束范式。

$$(1) (\forall x)(P(x) \rightarrow (\exists y)Q(x, y)) (5分)$$

$$(2) (\forall x)(\forall y)(\forall z)(P(x, y, z) \wedge ((\exists u)Q(x, u) \rightarrow (\exists w)Q(y, w))) (5分)$$

$$(3) (\exists x)P(x, y) \leftrightarrow (\forall z)Q(z) (5分)$$

$$(4) (\neg(\exists x)P(x) \vee (\forall y)Q(y)) \rightarrow (\forall z)R(z) (5分)$$

$$(5) (\forall x)(P(x) \rightarrow (\forall y)((P(y) \rightarrow (Q(x) \rightarrow Q(y))) \vee (\forall z)P(z))) (5分)$$

$$(6) (\forall x)(P(x) \rightarrow (\exists y)Q(x, y)) \vee (\forall z)R(z) \text{ (5分)}$$

$$(7) (\exists y)(\forall x)(\forall z)(\exists u)(\forall v)P(x, y, z, u, v) \text{ (5分)}$$

$$(1) \text{原式} = (\forall x)(\neg P(x) \vee (\exists y)Q(x, y))$$

$$= (\forall x)(\exists y)(\neg P(x) \vee Q(x, y))$$

$$(2) \text{原式} = (\forall x)(\forall y)(\forall z)(P(x, y, z) \wedge ((\forall y)\neg Q(x, y) \vee (\exists w)Q(y, w)))$$

$$(\forall x)(\forall y)(\forall z)(\forall u)(\exists w)(P(x, y, z) \wedge (\neg Q(x, u) \vee Q(y, w)))$$

$$(3) \text{原式} = ((\exists x)P(x, y) \vee \neg((\forall z)Q(z))) \wedge (\neg((\exists x)P(x, y)) \vee (\forall z)Q(z))$$

$$= ((\exists x)P(x, y) \vee (\exists z)\neg Q(z)) \wedge ((\forall x)\neg P(x, y) \vee (\forall z)Q(z))$$

$$= (\exists x)(\exists z)(P(x, y) \vee \neg Q(z)) \wedge (\forall x)(\forall z)(\neg P(x, y) \vee Q(z))$$

$$= (\exists x)(\exists z)(P(x, y) \vee \neg Q(z)) \wedge (\forall u)(\forall w)(\neg P(u, y) \vee Q(w))$$

$$= (\exists x)(\exists z)(\forall u)(\forall w)((P(x, y) \vee \neg Q(z)) \wedge (\neg P(u, y) \vee Q(w)))$$

$$(4) \neg(\neg(\exists x)P(x) \vee (\forall y)Q(y)) \vee (\forall z)R(z)$$

$$= ((\exists x)\neg P(x) \wedge (\exists y)\neg Q(y)) \vee (\forall z)R(z)$$

$$= (\exists x)(\exists y)(\forall z)((\neg P(x) \wedge \neg Q(y)) \vee R(z))$$

$$(5) \text{原式} = (\forall x)(\neg P(x) \vee (\forall y)((\neg P(y) \vee (\neg Q(x) \vee Q(y))) \vee (\forall z)P(z)))$$

$$= (\forall x)(\neg P(x) \vee (\forall y)((\neg P(y) \vee \neg Q(x) \vee Q(y))) \vee (\forall z)P(z))$$

$$= (\forall x)(\forall y)(\forall z)(\neg P(x) \vee \neg P(y) \vee \neg Q(x) \vee Q(y) \vee P(z))$$

$$(6) \text{原式} = (\forall x)(P(x) \rightarrow Q(x, f(x))) \vee (\forall z)R(z)$$

$$= (\forall x)(\forall z)(P(x) \rightarrow Q(x, f(x)) \vee R(z))$$

$$= (\forall x)(\forall z)(\neg P(x) \vee Q(x, f(x)) \vee R(z))$$

$$(7) \text{原式} = (\forall x)(\forall z)(\exists u)(\forall v)P(x, a, z, u, v)$$

$$= (\forall x)(\forall z)(\forall v)P(x, a, z, f(x, z), v)$$